



Figure 1: circleDoc module documentation index



Figure 2: circleDoc module documentation detailed

The following commands test the *simpleServer* server process:

```
1> c(simpleServer).
{ok,simpleServer}
2> simpleServer:start_link().
{ok,<0.40.0>}
3> simpleServer:get_count().
0
4> simpleServer:get_count().
1
5> simpleServer:get_count().
2
6> simpleServer:say_hi().
Hi!
ok
7> simpleServer:stop().
ok
24>
=INFO REPORT==== 27-May-2013::12:27:43 ===
terminating
8> simpleServer:say_hi().
ok
9> whereis(simpleServer).
undefined
```

The really interesting thing is that by writing Erlang code

to supervise the server, when the *simpleServer* server goes down for some reason, the supervisor will start another *simpleServer* process!

The OTP supervisor code (*simpleSup.erl*) is as follows:

%% Programmer: Mihalis Tsoukalos

%% Date: Monday 27 May 2013

```
-module(simpleSup).
-behaviour(supervisor).
-export([start_link/0]).
-export([init/1]).
-define(SERVER, ?MODULE).

start_link() ->
    supervisor:start_link({local, ?SERVER}, ?MODULE, []).

init([]) ->
    RestartStrategy = one_for_one,
    MaxRestarts = 1000,
    MaxSecondsBetweenRestarts = 3600,
    SupFlags = {RestartStrategy, MaxRestarts,
                MaxSecondsBetweenRestarts},
    Restart = permanent,
    Shutdown = 2000,
    Type = worker,
    AChild = {'simpleServer', {'simpleServer', start_link,
                               []},
              Restart, Shutdown, Type, [simpleServer]},
    {ok, {SupFlags, [AChild]}}.
```

In order to test the supervisor code, I ran the following commands in the Erlang shell:

```
1> c(simpleSup).
{ok,simpleSup}
2> {ok, Pid} = simpleSup:start_link().
{ok,<0.45.0>}
3> unlink(Pid).
true
4> simpleServer:start_link().
{error,[already_started,<0.46.0>]}
5> simpleServer:get_count().
0
6> simpleServer:get_count().
1
7> simpleServer:say_hi().
Hi!
ok
8> simpleServer:get_count().
3
9> gen_server:cast(simpleServer, crash).
=INFO REPORT==== 27-May-2013::13:02:04 ===
terminating
=ERROR REPORT==== 27-May-2013::13:02:04 ===
** Generic server simpleServer terminating
```